

Wireless Energy Transfer From Space to Ground via Satellite Constellation Grids

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Abstract—This letter presents a framework for space-to-ground wireless energy transfer (WET) for wirelessly chargeable devices (WCD) located in remote areas or disaster situations. We consider a grid of multi-antenna satellites that charge a WCD within line-of-sight. Closed-form expressions for harvested energy are derived considering maximum ratio transmission (MRT) ensuring that the WCD meets its circuit charging threshold P_{th} . Simulations elucidate that milli-joule-level energy can be harvested during satellite grid visibility, with charging efficiency influenced by the number of satellites, their altitude, charging frequency, and grid inclination. All codes used in this letter can be found from: https://github.com/moshehab570/Satellite_WET.git

Index Terms—Satellite constellations, wireless energy transfer, MIMO.

I. INTRODUCTION

IN DISASTER-AFFECTED or remote regions such as vast oceans and deserts, access to stable power supplies is often disrupted or entirely unavailable, severely limiting the operation of critical communication systems, sensors, and rescue equipment. Traditional power delivery methods—such as wired connections or manual battery replacement can be slow, dangerous, or logistically impossible to deploy in such conditions. WET emerges as a fast and reliable solution, enabling the remote powering or recharging of devices without physical contact or human presence.

Using WET, low-power wireless and electronic devices can be recharged remotely using terrestrial power beacons, aerial platforms, or satellites. For instance, in disaster situations such as earthquakes and floods, a wirelessly chargeable phone with a depleted battery could be wirelessly powered, allowing a victim to make emergency calls, share injury details, or help rescuers locate them under debris. Additionally, WET can reactivate sensors and cameras in power-limited areas to retrieve crucial data about past events [1].

To this end, terrestrial or aerial WET providers might be infeasible, costly, or slow to deploy in remote and disaster-

stricken areas. Nowadays, the availability of thousands of low Earth orbit (LEO) satellites plays an important role in providing connectivity to off-grid isolated regions and to civilians affected by disasters such as earthquakes, floods, and military conflicts. These scenarios can exploit the wide satellite coverage and near-global visibility to deliver milliwatt-level power, which is sufficient to operate low-power devices and to support user devices for short periods during emergencies. However, research on delivering energy from satellite constellations to devices in such conditions remains limited.

In this context, the work in [2] discussed an age-optimized simultaneous wireless information and power transfer for energy-limited devices. Their work provided useful insights on the age-energy tradeoff from queuing theory perspective, while providing solution based on meta-learning. However, it did not include closed-form solutions for the harvested energy or how the system would scale up with multiple satellites and antennas. The authors of [3] discussed stochastic geometry approaches for the analysis of dense LEO satellite systems. Meanwhile, [4] provided an overview of architecture and key technologies, including beam design for satellite constellations and the impact on system performance indicators such as coverage. Furthermore, [5] presented a detailed survey of MIMO satellite systems.

The authors of [6] proposed a novel distributed beamforming technology to enhance the space-to-ground link budget based on the superposition of electromagnetic waves radiated from multiple satellites. Meanwhile, the works in [7] and [8] discussed inter-satellite optical power transfer, and the authors of [9] illustrated a movable antenna setup to significantly reduce interference leakage.

To this end, we promote the idea of providing energy from space to low-power IoT or end-user devices located in remote and disaster areas, where access to power supplies might be challenging. We start by deriving the harvestable energy at a WCD within the charging visibility range of a grid of satellites equipped with multiple antennas. The obtained expressions account for the azimuth angle of inclination of the satellite orbit plane with respect to the WCD, the WCD circuit threshold, as well as the MRT gain of the multi-antennas at the satellite grid. We present extensive numerical results to affirm that it is possible to harvest a reasonable amount of energy during the visibility period of a satellite grid, specially for higher number of satellites and antennas. The results highlight the harvestable energy characterization for different satellite heights, carrier frequencies, WCD circuit threshold values, and satellites' azimuth inclination angles. Interestingly, a WCD can

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top of the this page. The details of the solution are depicted in Appendix. This formula further reduces to the following simpler expression for an azimuth angle $\phi = 0$ (i.e., when the satellite orbit and the WCD are in the same plan)

$$E_h(\phi = 0) = \mu \frac{\pi - 2 \tan^{-1} \left(\frac{H \cot(\frac{\omega T}{2})}{H+2R} \right)}{H^2 \omega + 2HR\omega}, \quad (14)$$

which acts as an upper-bound.

A special case of energy harvesting is to make use of the merits of beamforming via maximum ratio transmission [15] by considering M antennas per satellite. In such case, it is reasonable to assume that if the satellites are close to each other (i.e., satellite grid), then the channel coefficient between them and the ground device, would be almost constant due to the large distance between the satellites and the WCD. According to (10), as shown at the bottom of the page, and (13) in [15], the amount of harvested energy is multiplied by the square of the total number of antennas (MN^2) and the total amount of harvested energy at the earth device is theoretically upper bounded by

$$E_{h_u} = 2(MN^2)\mu \frac{\pi - 2 \tan^{-1} \left(\frac{H \cot(\frac{\omega T}{2})}{H+2R} \right)}{H^2 \omega + 2HR\omega}. \quad (15)$$

Remark 1: Interestingly, (15) reveals that increasing the number of satellites or the number of antennas per satellite have a similar effect on boosting the upper bound of the harvested energy. Thus, it is possible to reduce the number of satellites in a satellite grid by equipping the satellites with more antennas, which would save the costs of launching larger number of satellites per grid from an economic point of view. However, more studies are required on the practicality of antenna densification of satellites and the realistic performance in that case.

C. Charging Efficiency

Finally, defined as the ratio between the actual harvested energy and the available energy from the satellite grid during its visibility path (i.e., when $P_{th} = 0$), the charging efficiency $\eta_c = \frac{E_h}{E_h(P_{th}=0)}$ indicates the portion of energy that is actually harvested during the satellites visibility range due to the circuit threshold limitations.

D. Phase Misalignment

Taking into consideration the satellite phase shifts according to [16], the phase variation at the WCD could be modeled as $[e^{j\psi_1}, \dots, e^{j\psi_N}]$, where ψ_n denotes the phase error at satellite n . For i.i.d. phase errors with $\psi_n \sim \mathcal{N}(0, \sigma_\psi^2)$, the cross power terms are attenuated by the factor $e^{-\sigma_\psi^2}$, where σ_ψ^2 is the variance of the phase error distribution.

TABLE I
SIMULATION PARAMETERS

Parameter	Value	Parameter	Value
R	6378 Km	H	200 Km
G	$6.67 \times 10^{-11} \frac{m^3}{Kg \cdot s^2}$	M_e	5.97×10^{24} Kg
G_T	50 dB	G_r	10 dB
Ω	1.29	f	868 MHz
m	19.4	P_t	40 dBm
b_o	0.158	η_h	70%

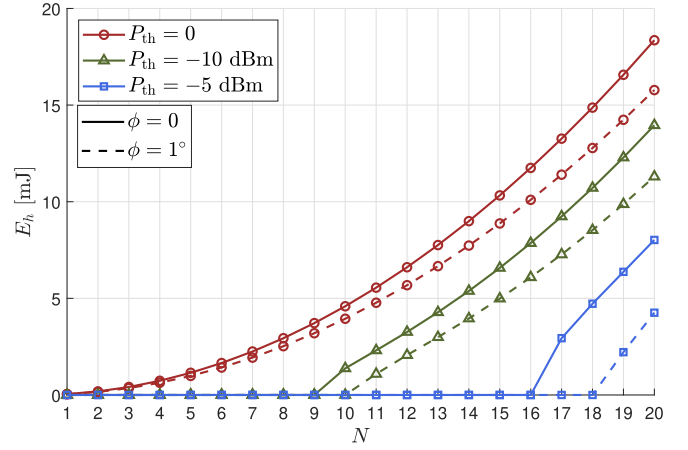


Fig. 2. Harvested energy vs the number of satellites for different WCD sensitivities.

III. NUMERICAL RESULTS

We consider a grid of LEO satellites operating within the L-Band (i.e., 858 MHz). Each satellite is equipped with four antenna elements, moving together while charging a WCD. Moreover, we consider that the charging window is determined by both the sensitivity of the WCD and the horizon point. For each value of P_{th} depicted in the following figures, we calculate d_c and θ_c , then T from (8), (9), (13), respectively. Note that, T spans the satellite charging path as shown in Fig. 1. Table I illustrates all the simulation parameters in all simulations unless stated otherwise. Note that the simulation parameters including channel parameters are empirically chosen within the practical ranges of LEO satellites operating in light shadowing as in well-established works such as [10]. Finally, the parameters of the WCD are selected based on the measured characteristics of the circuit in [17].

Fig. 2 depicts the harvested energy at the WCD as a function of the constellation size, under varying sensitivity thresholds and azimuth angle ϕ values. As anticipated from (15), increasing the number of satellites enhances the harvested energy. Apparently, when accounting for practical WCDs with non-ideal sensitivity, a minimum constellation size is required to ensure successful energy harvesting, e.g., 9 and 16 for $P_{th} = -10, -5$ dBm respectively, when $\phi = 0^\circ$. This is because increasing the sensitivity of the WCD decreases

$$E_h = \frac{2\mu \tan^{-1} \left[\frac{(H^2 + 2HR + 2R^2 + 2R(H+R) \cos(\phi)) \tan[\frac{\omega T}{2}]}{\sqrt{H^4 + 4H^3R + 6H^2R^2 + 4HR^3 + 2R^4 - 2R^2(H+R)^2 \cos[2\phi]}} \right]}{\omega \sqrt{H^4 + 4H^3R + 6H^2R^2 + 4HR^3 + 2R^4 - 2R^2(H+R)^2 \cos[2\phi]}} \quad (10)$$

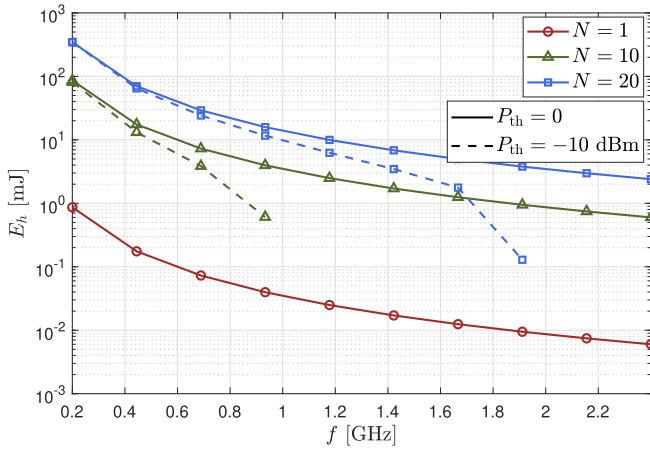


Fig. 3. Harvested energy vs the operating frequency for different number of satellites as well as practical and ideal WCD circuits.

the duration of the charging window and consequently the achievable harvested energy. Note that, $P_{th} = 0$ means that the circuit is able to sense and harvest energy even if the received power is too small (i.e, $P_r \rightarrow 0$). This is considered to be the benchmark ideal case. Also, observe that when the device is slightly offset (i.e, by only 1°) from satellites' ground track the achievable harvested energy significantly declines.

Fig. 3 shows how the harvested energy is affected by the operating frequency due to the higher free-space path losses for higher frequencies. The figure elucidates that this effect is more pronounced for smaller constellation sizes, whereas larger constellations enhance spatial diversity and thus improve the total harvested energy. Moreover, when accounting for the sensitivity of the WCD, the achievable harvested energy is further constrained, rendering certain portions of the frequency spectrum infeasible for energy harvesting depending on the number of satellites being used. For instance, the maximum feasible frequency for 10 satellites is 950 MHz, while it is extended to 1.9 GHz for 20 satellites.

Fig. 4 depicts how increasing the number of satellites can enable charging from higher distances. For practical WCDs with non-ideal sensitivity, the maximum allowable (i.e, feasible) constellation height becomes a function of the constellation size. For instance, as observed from the figure, the maximum feasible height for charging a WCD using a grid of 10 satellites is 220 km, whereas a grid of 20 satellites extends the feasible height to nearly 440 km. In the most favorable scenario, simultaneous charging by 20 at $d(t) = H = 200$ km yields a received power of approximately -3 dBm. This value is well-below the saturation point of the circuit in [17] which is why this operating region has not been considered in (7).

Fig. 5a illustrates how the charging efficiency is affected by the satellite azimuth angle ϕ . The main takeaway here is that, the charging efficiency is very sensitive to ϕ as it decreases rapidly and energy availability is not well exploited when the satellite plane is slightly inclined, specially for higher receiver sensitivity and lower number of satellites. Therefore, when multiple satellite swarms are available, the swarm whose ground track is closest to the device contributes dominantly to the harvested energy. Meanwhile, Fig. 5b illustrates the effect

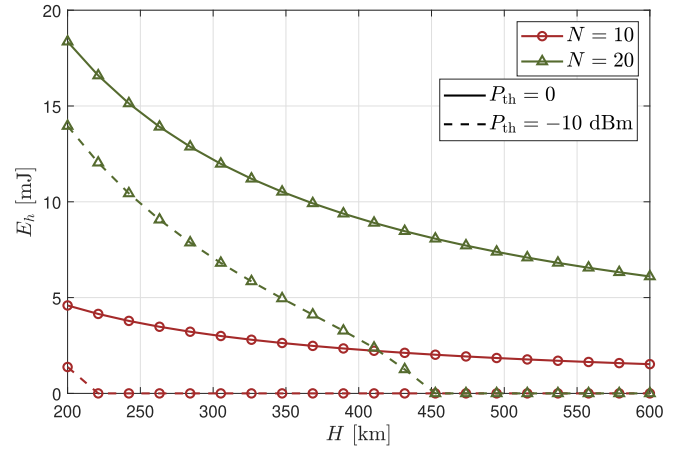
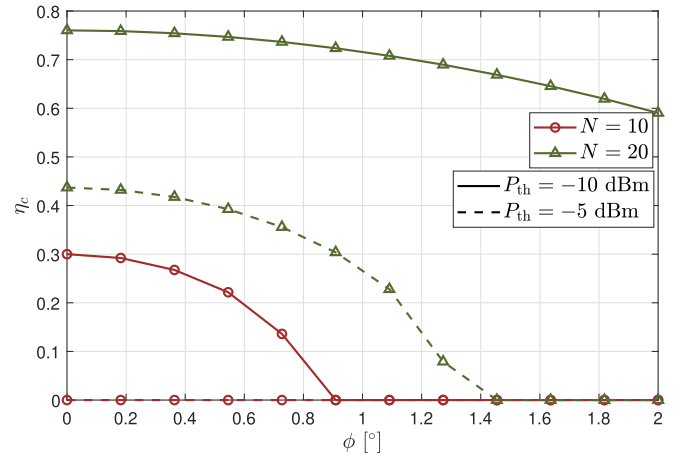
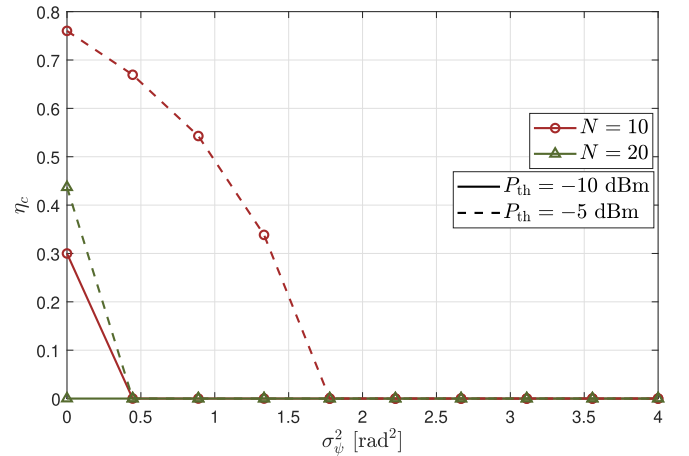


Fig. 4. Harvested energy vs the height of the swarm for different number of satellites and practical and ideal WCD circuits.



(a) Charging efficiency vs ϕ for different number of satellites.



(b) Charging efficiency vs σ_ϕ^2 for different number of satellites.

Fig. 5. Charging efficiency.

of phase misalignment on the charging efficiency. We observe that the charging efficiency significantly decays, when the variance of the phase misalignment σ_ϕ^2 becomes higher.

IV. CONCLUSION

This letter presented a framework for space-to-ground energy transmission using satellites, which could be applied

to charge IoT or user devices in remote areas or disaster situations. In summary, we suggested a layout, where a group of stacked satellites equipped with multiple antennas are charging a ground WCD, within the visibility range of its orbit. We derived closed form expressions for the amount of energy harvested aided by the maximum ratio transmission gain. The simulation results showed the interplay between the harvested energy, satellite height, number of satellites and transmission frequency for different circuit cut-off setups. Future research directions include but are not limited to exploring laser based charging from space including inter-satellite WET. Moreover, the model provided in this letter can be extended to joint energy and information transfer from satellites, for instance via modulating the noise statistical properties (i.e, noise modulation).

APPENDIX

EVALUATION OF THE INTEGRAL IN (11)

First, define the constants

$$A = (R + H)^2 + R^2 = H^2 + 2HR + 2R^2,$$

$$B = 2R(R + H) \cos \phi.$$

Then, the integral becomes

$$E_h = \mu \int_0^T \frac{1}{A - B \cos(\omega t)} dt. \quad (16)$$

From trigonometry, we have

$$\cos(\omega t) = \frac{1 - \tan^2\left(\frac{\omega t}{2}\right)}{1 + \tan^2\left(\frac{\omega t}{2}\right)}. \quad (17)$$

Let $u = \tan\left(\frac{\omega t}{2}\right)$. Differentiating, we get

$$dt = \frac{2}{\omega} \frac{du}{1 + u^2}. \quad (18)$$

Substituting into the integral yields

$$E_h = \frac{2\mu}{\omega} \int_0^T \frac{du}{A(1 + u^2) - B(1 - u^2)} \quad (19)$$

$$= \frac{2\mu}{\omega} \int_0^T \frac{du}{(A - B) + (A + B)u^2}. \quad (20)$$

According to [14], this integral has a standard form and can be evaluated as

$$\frac{2\mu}{\omega \sqrt{A^2 - B^2}} \tan^{-1} \left(\sqrt{\frac{A + B}{A - B}} u \right). \quad (21)$$

Applying the limits of the integral and expanding the constants yields the expression in (10).

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